MILITARY SPECIFICATION

TROUSERS, FIREMEN'S, STRUCTURAL/ALUMINIZED PROXIMITY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for a flame resistant structural firefighters trouser with a para-aramid and PBI blend cover and a waterproof moisture barrier lining. In addition the trouser has either a removable aluminized Para-aramid or a removable aluminized Para-aramid/PBI cover for proximity use.

1.2 Classification. The trouser covered by this specification shall be of the following types, classes, sizes, and lengths (see 6.2):

1.2.1 Types and classes.

Type I - Structural trouser

Class 1 - Trouser with neoprene coated lining
Class 2 - Trouser with PTFE laminate lining

Type II - Trouser cover (Aluminized)

1.2.2 Sizes. 34, 36, 38, 40, 42, 44 and 46.

1.2.3 Lengths. Regular and long.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Officer in Charge, Navy Clothing and Textile Research Facility, 21 Strathmore Road, Natick, MA 01760-2490 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8415

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2.1.1 Specifications, standards, and handbooks. The following specifications, standards and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

KK-L-2004	- Leather, Cattlehide, Deerskin and Horsehide, Chrome Tanned
NN-P-71	- Pallets, Material Handling, Wood,Stringer, Construction, 2 Way and 4 Way Entry (Partial)
ZZ-T-831	- Tubing; Rubber and Plastic
DDD-L-20	- Label; For Clothing, Equipage, and Tentage (General Use)

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MIL-W-5664	- Webbing, Textile, Elastic, Cotton
MIL-P-15011	- Pallet, Material Handling, Wood Post Construction, 4-Way Entry
MIL-B-17757	- Boxes, Shipping, Fiberboard, (Modular Sizes)
MIL-C-19699	- Cloth, Coated (Nylon Taffeta)
MIL-E-20652	- Eyelet, Metallic, Rolled Flange Type, and Eyelet Washer
MIL-F-21840	- Fastener Tapes, Hook and Pile, Synthetic
MIL-C-24929	- Cloth, Laminated, Aluminized (Para-Aramid Base)
MIL-C-24939	- Cloth, Ripstop, Para-Aramid/PBI, Water Repellent
MIL-C-24941	- Cloth, Laminated, Aluminized (Para-Aramid/PBI Base)
MIL-T-43636	- Thread, Aramid
MIL-C-43774	- Cloth, Plain or Pajama Check Weave, Aramid
MIL-T-44100	- Thread, Para-Aramid, Spun, Intermediate Modulus

STANDARDS

FEDERAL

FED-STD-191	- Textile Test Methods
FED-STD-311	- Leather, Methods of Sampling and Testing
FED-STD-751	- Stitches, Seams, and Stitchings

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MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	- Marking for Shippment and Storage
MIL-STD-147	- Palletized Unit Loads
MIL-STD-1666	- Classification of Defects for Evaluating Quality of Aluminized Firemen's Clothing
MS-27980	- Fastener, Snap, Style 2

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(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

LAWS AND REGULATIONS

U.S. POSTAL SERVICE MANUAL

(Copies of the manual may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issue of documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. *Unless otherwise specified, the* issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification

(Applications for copies should be addressed to the American Trucking Association, ATTN: Traffic Department, 1616 P Street, N.W., Washington, DC 20036-1404).

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, Illinois 60606-5808).

JOINT COMMITTEE ON PRINTING (JCP) PUBLICATIONS

Government Paper Specification Standards

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402).

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2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Guide sample. Samples, when furnished, are solely for guidance and information to the contractor (see 6.3). Variations from this specification may appear in the sample in which case this specification shall govern.

3.2 First article. When specified, the contractor shall furnish sample unit(s) for first article inspection and approval (see 4.3 and 6.2).

3.3 Materials.

3.3.1 Basic material (structural trousers). The basic material for the structural trouser shall be a water repellent para-aramid/PBI ripstop fabric conforming to MIL-C-24939.

3.3.2 Basic material (Aluminized cover). The basic material for the aluminized cover shall be either of the following fabrics. A highly reflective aluminum surface laminated to a knitted para-aramid base fabric conforming to Type II of MIL-C-24929 or a highly reflective aluminum surface laminated to a knitted para-aramid/PBI base fabric conforming to MIL-C-24941.

3.3.3 Lining (Class 1 trouser). The material for lining the class 1 structural trouser shall be a coated nylon taffeta conforming to Type II of MIL-C-19699.

3.3.4 Lining (Class 2 trouser). The material for lining the class 2 structural trouser shall be a two layer laminated cloth which is waterproof and moisture vapor permeable. The face of the fabric shall be a flame resistent, aramid pajama check weave cloth conforming to MIL-C-43774, color natural. The back side of the laminated cloth shall be a micropourous, expanded polytetraflourothylene film weighing 0.7 $\pm$ 0.3 oz/yd^2 as tested in 4.4.1 and an adhesive suitable for the purpose intended and compatible with the film. The face and back cloths shall be laminated together in such a manner that the assembled (finished) cloth shall comply with the requirements specified in Table I when tested as specified in 4.4.1 (see 6.5).

TABLE I

Characteristic	Requirement
Moisture Vapor Transmission rate, Grams/m^2/24 hrs (min)	450
Hydrostatic Resistance, PSI (min)	50
Water Permeability After Synthetic Perspiration	No leakage

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3.3.5 Thread. The thread for assembling the trouser shall be a non-melting nylon (aramid), natural shade, conforming to Size B of MIL-T-43636. As an alternate, a thread conforming to Size T-50 of MIL-T-44100, natural in color may be used.

3.3.6 Leather. The leather for the protective knee patch, leg tabs and for binding the leg bottoms shall be a flesh split cattlehide, natural color of tannage, of not less than 2-3/4 nor more than 3-1/2 ounce thickness conforming to type IV, class 2 of KK-L-2004 after silicone treatment specified in 3.3.6.1, except for the following;

a. Shrinkage temperature shall be not less than 105°C.
b. Tests for percent ash and chloroform-soluble material not required.

3.3.6.1 Silicone treatment. The leather shall be treated with an approved durable silicone compound (see 6.9) to lower its water water absorption. The treated leather shall not absorp more than 30 percent water when tested as specified in 4.4.1.

3.3.7 Reflective tape. The reflective tape shall be in accordance with 3M style 8487 flourescent lime/yellow, 3 inch wide or equal (see 6.6 and 6.10).

3.3.8 Elastic tubing and connectors.

3.3.8.1 Elastic tubing. The seamless tubing used in the leg protector shall conform to types I or V, classes 2 or 3 of ZZ-T-831. Any color of tubing may be used.

3.3.8.2 Tubing connector. The tubing connectors used to join the two ends of the latex tubing shall be straight nylon hose fittings in accordance with McMaster Carr Supply Co. style # 5380-K12 or equal (see 6.7 and 6.10).

3.3.9 Hook and pile fastener tape. The hook and pile fastener tapes shall conform to Type II, Class 1 of MIL-F-21840 and shall be 1 inch and 2 inch wide. The color shall be natural, white or black.

3.3.10. Eyelets and washers. The eyelets and washers for the pocket drain holes shall conform to rolled flange type, M20652/1-BBE114 and M20652/1-BBW101 of MIL-E-20652.

3.3.11 Suspender assembly.

3.3.11.1 Buckle. The front of the clamp type buckle shall be made from 0.020 inch, 3 temper, SAE 1010 cold-rolled steel. The back of the buckle shall be made of 0.058 inch soft basic steel wire. The buckle shall be zinc plated and lacquered black or shall have a nickel or chrome finish (see Figure 3).

3.3.11.2 Slide. The slide for the suspender strap shall be die cast zinc and shall be lacquered black or shall have a chrome or nickel finish. It shall be 0.125 inch thick with two 1-1/2 by 3/16 inch openings (slots). Overall dimension shall be 1-3/4 by 13/16 inch (see 6.4).

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3.3.11.3 Webbing. The elastic webbing shall be 1-1/2 inches wide conforming to type I, class 2 of MIL-W-5664. The color of the webbing shall be natural, white, black or olive drab.

3.3.12 Snap fasteners.

3.3.12.1 Snap fasteners (standard). The snap fasteners for attaching the snap-in liner and for attaching the aluminized cover with the exception of the left fly socket stud combination, shall consist of a button and socket (female half) and a stud and eyelet (male half). The fasteners shall be style 2, 24-line (size 1), nickel plate or bright chrome finish conforming to MS-27980 and the following part numbers:

Button - 1N or 1C	Socket - 6N or 6C
Stud - 7N or 7C	Eyelet - 8N or 8C

3.3.12.2 Snap fasteners (special). The combination stud and socket fastener used on the left fly shall be manufactured of nickel plated brass in accordance with Scovill #393 stud and #588 socket or equal (see 6.8 and 6.10).

3.3.13 Labels.

3.3.13.1 Label, Structural trouser. Each pair of trousers shall have a combination size, instruction and identification label conforming to Type VI, Class 14 of *DDD-L-20*, except that the colorfastness requirements shall not apply (see 6.12). The label shall bear the following inscription:

TROUSERS, FIREMEN'S, STRUCTURAL (CLASS 1) (EXAMPLE)
FIBER CONTENT:
Outershell - 60% Para-aramid/40% PBI
Lining - Nylon taffeta (EXAMPLE)
CONTRACT NO: DLA-100-00-0-0000 (EXAMPLE)
STOCK NO: 8415-00-000-0000 (EXAMPLE)
SIZE/LENGTH: MEDIUM/REGULAR (EXAMPLE)
NAME OF CONTRACTOR:
NAME OF MANUFACTURERE: (If other than contractor)

LAUNDERING INSTRUCTIONS
HAND WASH IN LUKE WARM WATER
DO NOT BLEACH
HANG TO DRY
DO NOT MACHINE WASH OR DRY CLEAN

WARNING

TROUSER MUST BE WORN WITH LINER INTACT AND ALL CLOSURES FASTENED WHEN IN USE. DO NOT KEEP IN DIRECT CONTACT WITH FLAME. DO NOT STORE THIS GARMENT IN DIRECT SUNLIGHT. KEEP GARMENT CLEAN - SOILING WILL REDUCE PROTECTIVE QUALITIES. FAILURE TO COMPLY WITH THESE INSTRUCTIONS MAY RESULT IN SERIOUS INJURY OR DEATH.

DO NOT REMOVE THIS LABEL

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3.3.13.2 Warning label, structural trouser. Each pair of trousers shall have a warning label conforming to Type IV, Class 9 of DDD-L-20 applied to the inside back lining (see 6.12). The label shall be applied to the lining by a red marking medium. The label shall be printed with block sansserif type with letters 3/4 to 1 inch in height. The label shall contain the following inscription:

WARNING
MUST BE WORN WITH
LINER IN PLACE

3.3.13.3 Label, proximity cover. Each proximity cover shall have a combination size, instruction and identification label conforming to Type VI, Class 14 of DDD-L-20 except that the colorfastness requirements shall not apply. The label shall bear the following inscription:

COVER, FIREMEN'S, TROUSERS,(PROXIMITY)
FIBER CONTENT:
Outershell - 100% Para-aramid (OR) 60% Para-Aramid/40% PBI
CONTRACT NO: DLA-100-00-0-0000 (EXAMPLE)
STOCK NO: 8415-00-000-0000 (EXAMPLE)
SIZE/LENGTH: MEDIUM/REGULAR (EXAMPLE)
NAME OF CONTRACTOR:
NAME OF MANUFACTURER (If other than contractor)

INSTRUCTIONS

1. CAUTION: MUST BE USED ONLY AS AN OUTER SHELL WITH STRUCTURAL TROUSER.
2. Hang to dry and store properly to prevent damage to aluminized surfaces.
3. Clean by gently rubbing surface with cloth using mild soap, PERCHLORETHYLENE or ISOPROPANOL. When cleaning with perchlorethylene or isoproponol, ensure that you have adequate ventilation and that you have proper gloves and eye protection.
4. To maintain reflective properties and to extend service life, use proximity cover for intended purpose only.
5. Do not use aluminized paint to repair worn out surface as it will not provide protection.

NOTE: Aluminized coating is damaged by contact with fire extinguishing agent Potassium Bicarbonate Powder (PKP). Before storing garment with PKP residue present, the following is recommended.

1. Dry powder should be removed with dry brush or blower.
2. Wet powder should be removed by thoroughly flushing with water and wiping the surface down with a clean soft cloth. Damp or wet garments should be hung to dry.

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3.3.13.4 Warning label, proximity cover. Each proximity cover shall have a warning label conforming to Type IV, Class 9 of DDD-L-20 applied to the inside back lining. The label shall be applied to the lining by a red marking medium. The label shall be printed with block sansserif type with letters 3/4 to 1 inch in height. The label shall contain the following inscription:

WARNING
MUST BE WORN WITH
STRUCTURAL TROUSERS

3.3.14 Instruction Manual. An instruction manual shall be furnished by the contractor with each trouser. The manual shall be inserted into the polyethylene bag during packaging, and placed so that it is visible (see 5.1.1.1). The printing, collation, folding and stitching of the manual shall be in accordance with the standard sample (see 6.3). The cover shall be antique white book paper conforming to Joint Committee Printing-A-100, 140 pound basis weight 25 X 38 inches per 1000 sheets. The text paper shall be white offset book paper conforming to JCP-A-60, 120 pound basis weight 25 X 38 inches per 1000 sheets. A sample of the manual plus four glossy pictures shall be obtained from the contracting officer.

3.3.15 Repair of aluminum film. The aluminum film used for patches for repairing uncoated or cracked areas on the finished cover shall be the same as specified for coating the base cloth. The area to be repaired shall not exceed 1 inch in the longest direction. The patch shall be completely bonded onto the aluminum film of the outer shell and shall be so placed that it extends not less than 1/2 inch beyond all edges of the uncoated or cracked area. No outer shell shall contain more than 3 patches. The unsupported aluminum film and self-curing adhesive used for repairing shall be obtained from the basic material supplier (see 3.3.2). The size of the patch shall not exceed 2 inches in any direction.

3.4 Design. The trousers shall be of a two piece design, an inner structural trouser fabricated from a para-aramid/PBI fabric and lined with a moisture proof barrier fabric and an outer proximity cover fabricated from a highly reflective aluminized coated para-aramid/PBI blend fabric. The structural trousers have a fastener tape fly closure and elastic suspender straps stitched to the waistband at front and back. The knees are protected by leather patches and each leg has a strip of reflective tape. The right leg has a cargo pocket and flap. The outer cover is designed to fit over the structural trousers. Attachment is made by snap fasteners at the waist and the fly is closed by means of fastener tape. The inside bottom of each leg is fitted with an elasticized protector (see figures 1, 2 and 3).

3.4.1 Figures. Figures are furnished for informational purposes only. When inconsistencies exist between the written specification and the figures, the written specification shall govern.

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3.5 Patterns. Standard patterns to be used to cut working patterns will be furnished by the Government (see 6.3). The patterns allow a 1/2 inch seam allowance for major joining seams, except as otherwise indicated in Tables II and III. The working patterns shall be identical to the Government patterns. Neither the Government patterns nor the working patterns shall be altered in any way except that additional notches for use during construction are allowed on the working patterns.

3.5.1 Pattern parts. The component parts of the trousers shall be cut from the materials specified and in accordance with the number of parts required for the manufacturing process as follows:

MATERIAL	NOMENCLATURE OF PATTERN PARTS	CUT PARTS
	STRUCTURAL TROUSERS (TYPE I)	
Base fabric	Front	2
	Back	2
	Waistband facing	4
	Left fly	1
	Right fly	2
	Cargo pocket (all sizes)	1
	Cargo pocket flap (all sizes)	1
Moisture barrier	Front	2
	Back	2
	Cargo pocket reinforcement (all sizes)	1
	Cargo pocket flap reinforcement (all sizes)	1
Templates	Marker for back suspender location (all sizes)	-
	Marker for waistband studs (for attaching aluminized cover)	-
	ALUMINIZED TROUSER COVER (TYPE II)	
Aluminized	Front	2
	Back	2
	Waistband (Facing)	2
	Left fly extension	1
Leather	Knee patch	2
Moisture barrier	Leg protector	2
Template	Marker for waistband snaps (for attaching cover to structural trousers)	-
	Marker for right fly opening fastener tape location (all sizes)	-

3.6 Construction.

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3.6.1 Stitches, seams and stitchings. Stitch, seam and stitching types specified in Tables II and III shall conform to FED-STD-751. Whenever two or more methods for seam or stitch types are given for the same part of the operation any one may be used. Seam allowance shall be maintained with seams sewn so that no raw edges, runoffs, twists, pleats, puckers or open seams result. All seams shall start and finish evenly. Thread tension shall be maintained so that there is no tight or loose stitching. Where stitch type 401 is used, the looper (underthread) shall be on the inside of the garment. Top stitching and edge stitching shall be 1/16 to 1/8 inch from the edge, hook and pile fastener tape and reflective tape stitching shall be 1/16 to 1/8 inch from the tape edge. The bight of the overedge stitching shall be 3/16 to 1/4 inch. A knife shall be used on the overedge machine with the guide set so that only the ravelled edges are trimmed. In order to minimize the size of the needle hole in the aluminized fabric, the needle used in the seaming operations shall be the smallest size which will accommadate the specified size thread.

3.6.2 Thread breaks and end of seams. End of all seams and stitching when not caught in other seams or stitching shall be backtacked not less than 3/8 inch. The ends of a continuous line of stitching shall be overlapped not less than 1/2 inch. Thread breaks (all stitch type) shall be repaired by stitching back of the break not less than 1/2 inch. Skipped stitches or thread breaks on 401 stitch type may be repaired by using 301 stitch type. When ends of overedge stitching are not caught in other seams or stitching, the ends shall be tacked. When stitch type 401 is permitted as an option, it shall not be used unless the ends are caught in other seams of stitching.

3.6.3 Stitches per inch. The minimum and maximum number of stitches per inch shall be as specified in Tables II and III.

3.7 Manufacturing operations requirements. The trousers shall be manufactured in accordance with the operation requirements specified in Tables II and III. The supplier is not required to follow the exact sequence of operations listed provided that the finish trousers are identical to those produced by following the sequence of operations listed in Tables II and III.

3.7.1 Size marking. The component parts of the trousers shall be bundled, marked or ticketed to assure proper size assembly. The use of metal fastening devices or sewn-on tickets for marking is prohibited. Adhesive tickets, when used, shall not discolor the material and the adhesive mass shall not adhere to the material upon removal of the tickets.

NOTE: The use of ink pad numbering machines, rubber stamp or pencil is allowed, provided the numbering does not show on the outside of the trousers and wherever possible, is covered by the seam allowance.

3.7.2 Abbreviations in table of operations. The abbeviations used in tables II and III are as follows:

Stch	-	Stitch
in	-	inch
Ndl	-	Needle
Bob	-	Bobbin
Lpr	-	Looper

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TABLE II - CONSTRUCTION OF STRUCTURAL TROUSERS

No.	Operation	Stch Type	Seam/ Stch type	Stch in	Thread Ndl Bob/ Lpr
1.	Cutting.				
	a. Cut the structural trousers in strict accordance with patterns which show directional lines, marks and notches for proper assembly. Directional lines shall not appear on the fabrics. Drill holes are not permitted.				
	b. The component parts of the base fabric may be cut from ends and when so cut shall approximate the color of the major cut parts.				
	c. The component parts of the lining may be cut from ends.				
	d. Cut leather leg tabs 1 inch wide by 4-1/2 to 5 inches long.				
	e. The face side of the materials to be assembled shall be considered as follows: Outershell - as marked Lining - uncoated side				
	NOTE: During the manufacturing of this garment, the water repellent coated side of the lining shall face away from the wearer in the finished trouser.				
2.	Replacement of defective components.				
	During the spreading, cutting and manufacturing operations, components having material defects or damages that are classified as damages in MIL-STD-1666 and section 4 shall be removed and replaced with nondefective and properly matched components.				

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TABLE II - CONSTRUCTION OF STRUCTURAL TROUSERS

No.	Operation	Stch Type	Seam/ Stch type	Stch in	Thread Ndl	Thread Bob/ Lpr
3.	Seam direction.					
	The lapping of the joining seams shall be as follows: 1. Fronts over back at side seams and inseams. 2. Left over right at seat seam and crotch seam.					
4.	Make waistband facing.					
	a. Position waistband, face to face, at center back and stitch 3/8 to 1/2 inch from the raw edge. Spread seam allowance open and flat. NOTE: For both inner and outer plies of waistband.	301	SSa-1	6-8	B	B
	b. Position inner and outer plies of waistband, face to face, and stitch 3/16 to 1/4 inch from raw edge, the ends and bottom edge of waistband.	301	SSe-2(a)	6-8	B	B
	c. Turn waistband to finished position, work out corners and edges and topstitch turned edges.	301	SSe-2(b)	6-8	B	B
	d. For ease in manufacturing, the raw edges may be stitched together, 1/4 inch from the raw edge.	301	SSa-1	6-8	B	B
	e. Center and attach nine snap fastener sockets and caps to the lower section of the facing as indicated by the pattern marks (with sockets facing away from wearer as worn).					
5.	Make right fly.					
	a. Overedge stitch the back straight edge of the underply of the right fly piece.	515 or 516 or 519	SSa-2	6-8	B	B

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TABLE II - CONSTRUCTION OF STRUCTURAL TROUSERS

No.	Operation	Stch Type	Seam/ Stch type	Stch in	Thread Ndl	Thread Bob/ Lpr
5.	Make right fly. (Cont'd)					
	b. Position a strip of 1 inch wide hook fastener tape to the face of the outerply of the fly piece as indicated by marks on patterns. Stitch along all edges of the tape.	301 or 401	LSbj-1	8-8	B	B
	c. Position the two plies of the right fly piece together, face to face, and stitch only the front curved edge of fly piece 3/16 to 1/4 inch from the raw edge.	301	SSa-1	6-8	B	B
6.	Prepare left fly piece.					
	a. Overedge stitch the back curved edge of left facing.	515 or 516 or 519	EFd-2	6-8	B	B
	b. Position a strip of 1 inch pile fastener tape to the face of the left fly piece as indicated by marks on patterns. Stitch along all edges of the tape.	301	LSbj-1	6-8	B	B
		301	SSa-1	6-8	B	B
7.	Make cargo pocket.					
	a. Position cargo pocket reinforcement to back side of cargo pocket. The periphery may be stay stitched.					
	b. Position three metal eyelets on cargo pocket in accordance with marks on patterns and attach with the center of each eyelet 1 (± 1/8) inch from the finished bottom edge of the pocket.					
	c. Turn under the top raw edge of the pocket 3/8 to 1/2 inch and fold top as indicated by marks on patterns forming a 3 inch hem. Stitch 1/8 to 3/16 inch and again 2-5/8 to 2-3/4 inch from top folded edge.	301	EFt-2	6-8	B	B

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TABLE II - CONSTRUCTION OF STRUCTURAL TROUSERS

No.	Operation	Stch Type	Seam/ Stch type	Stch in	Thread Ndl	Thread Bob/ Lpr
7.	Make cargo pocket. (Cont'd)					
	d. Position a 2 by 3 inch piece of hook fastener tape on the cargo pocket as indicated by marks on patterns. Stitch along all edges.	301	LSbj-2	6-8	B	B
	e. Fold front edge of cargo pocket on fold line closest to the raw edge, face to face, and stitch 1/8 to 3/16 inch from folded edge.	301	OSf-1	6-8	B	B
	f. Fold edge of cargo pocket on second fold line, back to back, and stitch 1/8 to 3/16 inch from folded edge.	301	OSf-1	6-8	B	B
8.	Make cargo pocket flap.					
	a. Position a 2 by 3 inch piece of pile tape on the underside of the cargo pocket flap as indicated by marks on patterns. Stitch along all edges.	301	LSbj-1	6-8	B	B
	b. Position the cargo pocket reinforcement on the inside of the top layer of the cargo pocket flap and edgestitch together along both side edges.	301	SSa-1	6-8	B	B
	c. Fold cargo pocket flap in half, face to face, as indicated by marks on patterns and stitch 3/8 to 1/2 inch from the raw side edges.	301	SSe-2(a)	6-8	B	B
	d. Trim points, turn cargo pocket flap to finished position and completly force out corners. Stitch the turned in sides and bottom folded edge 1/8 to 3/16 inch from the turned or folded edge.	301	SSe-2(b)	6-8	B	B

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TABLE II - CONSTRUCTION OF STRUCTURAL TROUSERS

No.	Operation	Stch Type	Seam/ Stch type	Stch in	Thread Ndl	Thread Bob/ Lpr
9.	Attach reflective tape and knee patches to legs.					
	a. Position reflective tape to lower legs in accordance with marks on patterns. The top edge of tape shall be positioned at the marks. The tape shall be parallel to the bottom edges of the legs. Stitch each side of the reflective tape to the legs.	301 or 401	LSbj-1	6-8	B	B
	b. Position leather knee patches to fronts of legs as indicated by marks on patterns. Double stitch 1/4 inch guage with outer row of stitching 1/8 to 3/16 inch from the raw edge of the leather patch.	301 or 401	LSbj-2	6-8	B	B
10.	Make suspender strap assembly.					
	a. The cut length of elastic suspender straps shall be 43 (± 1/4) inches for all sizes.					
	b. The cut length of the front attachment straps shall be 4-1/2 (± 1/4) inches.					
	c. Thread strap through wire part of clamp type buckle for a distance of 1-1/4 (+1/8) inches and stitch full width of strap with 3/4 to 1 inch boxstitch.	301	Boxstitch	6-8	B	B
	d. Thread free end of strap through slide and back through buckle.					
	e. The front attachment strap shall be threaded through other side of the slide then turned onto itself and stay stitched.					
	f. Cross the straps at a distance of 7-1/2 (+ 1/2, - 0) inches from the raw free end of the straps and join with a diamond stitch as shown in Figure 3. Outer row of stitching shall be 1/8 inch from edges of webbing.	301	SSa-1	6-8	B	B
	g. Reinforce the two ends of the diamond stitch with 1/2 to 5/8 inch horizontal bartack as shown in Figure 3.	301	Bartack	26-30 per bartack	B	B

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TABLE II - CONSTRUCTION OF STRUCTURAL TROUSERS

No.	Operation	Stch Type	Seam/ Stch type	Stch in	Thread Ndl	Thread Bob/ Lpr
11.	Prepare lining assembly and attach labels.					
	a. Stitch 3/8 to 1/2 inch from raw edge, the seat seam, crotch seam, outseam and inseam, face to face.	301 or 401	LSq-2(a)	6-8	B	B
	b. Turn one ply back sharply over the first row of stitching and stitch 1/8 to 3/16 inch from the folded edge.	301	LSq-2(b)	6-8	B	B
	or					
	c. As an alternate, operations 11.a and 11.b may be performed in one operation with a double lapped doubled stitched seam, 1/4 inch guage.	301 or 401	LSc-2	6-8	B	B
	d. Position and attach a combination label (see 3.3.13.1) to center back of lining to finish 1 to 2 inches below finished bottom edge of waistband.					
	e. Apply warning label, (see 3.3.13.2) horizontally across center back 1 to 2 inches below bottom edge of combination label.					
12.	Prepare outershell and attach pockets and flaps.					
	a. Join outseams 1/2 inch from the raw edge catching the ends of the reflective tape. Turn seam face side up and topstitch 1/8 to 3/16 inch from the turned edge.	301 or 401	LSq-2(a&b)	6-8	B	B
	b. Turn raw pleated front edge of cargo pocket under 3/8 to 1/2 inch. Lay cargo pocket flat with pleats closed. Turn bottom and back raw edge under 3/8 to 1/2 inch and position on right leg as indicated by marks on pattern. Topstitch side and bottom edges 1/8 to 3/16 inch from the turned edge. Place a second row of topstitching along bottom and back edge 1/4 inch inside of first topstitch.	301 or 401	LSd-1 and LSd-2	6-8	B	B

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TABLE II - CONSTRUCTION OF STRUCTURAL TROUSERS

No.	Operation	Stch Type	Seam/ Stch type	Stch in	Thread Ndl	Thread Bob/ Lpr
12.	Prepare outershell and attach pockets and flaps. (Cont'd)					
	c. The finished cargo pocket for all sizes shall measure 10 (± 3/8) inches deep at center. The width of the pocket shall be 8 (± 3/8) inches wide when measured across top edge of pocket from turned edge of bellows to back turned edge.					
	NOTE: Care shall be exercised so as not to catch the outer pleated edge of pocket in the stitching of the pocket front edge.					
	d. Turn under 1/4 to 3/8 inch, the top raw edge of the cargo pocket flap and double stitch 1/4 to 5/16 inch guage to right leg as indicated by marks on patterns. The inner row of topstitching shall be 1/4 (± 1/8) inch below top folded edge of the pocket.	301 or 401	LSba-2	6-8	B	B
	or					
	e. As an alternate, position the cargo pocket flap face down on leg, with raw edges facing towards the bottom of the leg and 3/8 inch below the mark on the patterns. Stitch 1/4 to 5/16 inch from the raw edge. Turn flap to finished position and top stitch 3/8 to 1/2 inch from turned edge of flap.	301 or 401	LSbl-2	6-8	B	B
	f. Finished cargo pocket flaps for all sizes shall measure 3-3/4 (± 1/4) inches wide at center and 8 (± 1/4) inches long when measured along bottom edge.					
	g. Stitch a 5/8 inch vertical bartack at the top corners of the cargo pocket, a 5/8 inch horizontal bartack at the bottom front corner of the cargo pocket and a 5/8 inch horizontal bartack at each top corner of the cargo pocket flap. The bartacks shall be superimposed on the previous line of stitching.	Bartack	--	42 per bartack	B	B

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TABLE II - CONSTRUCTION OF STRUCTURAL TROUSERS

No.	Operation	Stch Type	Seam/ Stch type	Stch in	Thread Ndl	Thread Bob/ Lpr
12.	Prepare outershell and attach pockets and flaps. (Cont'd)					
	h. Join 1/2 inch from the raw edges the crotch seam, seat seam and inseam catching the ends of the reflective tape. Turn seams to finished position and stitch 1/8 to 3/16 inch from the turned edge.	301	LSq-2(a&b)	6-8	B	B
	i. As an alternate, operations 12.a. and 12.h. may be performed using seam type LSb-2.					
13.	Join shell and lining.					
	Insert lining into outershell with coated side of lining away from the body as worn. Stitch around top and fly edge 3/8 to 1/2 inch from the raw edge.	301	SSa-1	6-8	B	B
14.	Attach waistband and fly pieces.					
	a. Position waistband facing to top edge of trousers, face to face, and stitch 1/4 to 3/8 inch from the raw edge the entire length of waistband catching the front and rear suspender straps in the position indicated by the pattern marks. Top edge of waistband shall finish with 1 to 1-1/4 inch of suspender strap inside of waist.	301	SSe-2(a)	6-8	B	B
	b. Position raw edge of left fly piece, face to face, to left fly raw edge with left end of waistband facing sandwiched between. Stitch top and front edge of fly piece 1/4 to 3/8 inch from the raw edge, the length indicated on pattern piece.	301	SSe-2(a)	6-8	B	B
	c. Turn left fly facing to finished position forcing out the corner and stitch back edge to left front 3/8 inch from overedged back edge.	301	Similar to LSag-3	6-8	B	B

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TABLE II - CONSTRUCTION OF STRUCTURAL TROUSERS

No.	Operation	Stch Type	Seam/ Stch type	Stch in	Thread Ndl	Thread Bob/ Lpr
14.	Attach waistband and fly pieces. (Cont'd)	301	SSe-2(a)	6-8	B	B
	d. Position the raw edge of top ply of right fly piece, face to face, to the right fly raw edge. Stitch edge of fly piece 1/4 to 3/8 inch from raw edge, the length indicated on patterns.					
	e. Stitch across raw top edges of right fly piece, face to face, 3/16 to 1/4 inch from raw edge.	301	SSe-2(a)	6-8	B	B
	f. Turn right fly piece to finished position and top stitch 1/8 to 3/16 inch, the turned edges.	301	SSe-2(b)	6-8	B	B
	g. Topstitch 1/8 to 3/16 inch from turned edge, the back edge of right fly piece.	301	LSs-2(b)	6-8	B	B
	h. With waistband facing and front and rear suspenders straps in finished position topstitch across top edge of trousers and front edges of left and right fly pieces 1/4 to 3/16 inch from the turned edges. Make certain that top stitching of left fly continues onto crotch closing seam.	301	SSe-2(b)	6-8	B	B
	i. Stitch down waistband facing through all plies, 1-3/4 (± 1/8) inch from top. finished edge of trousers. Stitching shall extend from left front edge to beginning of right fly piece.	301	Similar to LSag-3	6-8	B	B
	j. With front edges flat and smooth place a 1/2 to 5/8 inch horizontal bartack at base of fly securing both right and left fly pieces.	301	Bartack	26-30 per bartack	B	B
15.	Finish off top of trousers.					
	a. Stitch with a 1-1/8 (± 1/16) inch slanted boxstitch, the rear suspender straps through all plies.	301	Boxstitch	6-8	B	B
	b. Stitch with a 1-1/8 (± 1/16) inch boxstitch, the front suspender attachment strap through all plies.	301	Boxstitch	6-8	B	B

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TABLE II - CONSTRUCTION OF STRUCTURAL TROUSERS

No.	Operation	Stch Type	Seam/ Stch type	Stch in	Thread Ndl	Thread Bob/ Lpr
15.	Finish off top of trousers. (Cont'd)					
	c. The finished attached length of each strap shall be 40 (± 1) inches when measured along center of strap from the top folded edge of the waist to the outer folded end with buckle clamped in its maximum extended position.					
	d. The front attachment strap shall finish 3/8 to 1/2 inch above top edge of waistband.					
	e. Mark position of six snap fasteners on outside of waistband in accordance with marker template. Template shall be lined up with side seam and front edge (not right fly extension).					
	f. Attach a snap fastener stud and socket combination to left front edge of waistband as previously marked in 15.e. Stud shall face away from wearer as worn.					
	g. Attach five snap fastener stud and eyelet combinations to rest of previously marked waistband positions. Studs shall face away from wearer as worn.					
	h. Position and attach a snap fastener stud and eyelet combination to right front fly extension to correspond to socket in operation 15.f., when the fly is closed flat and smooth. Stud shall face away from wearer as worn					
16.	Finish bottom edge of trousers.					
	a. Turn under entire bottom edge of leg 3/8 to 1/2 inch and form a 1-1/2 (± 1/8) inch wide hem. Stitch top and bottom edges of hem 1/8 to 3/16 inch from each turned edge. Seams of lining and trouser outershell shall match up.	301	Similar to EFq-2	6-8	B	B

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TABLE II - CONSTRUCTION OF STRUCTURAL TROUSERS

No.	Operation	Stch Type	Seam/ Stch type	Stch in	Thread Ndl	Thread Bob/ Lpr
16.	Finish bottom edge of trousers. (Cont'd)					
	b. Center and attach a snap fastener socket and cap to the end of the leather leg tab. The center of the socket shall be 1/2 inch from the end of the leather tab.					
	c. Position to each inside leg seam, a leg tab with the socket end towards top of trouser and socket facing outwards. Bartack or tack to trouser leg through all plies full width of tab 1/2 to 5/8 inch from end of tab and 8-1/4 (± 1/4) inch from finished bottom edge of leg.	301 or Bartack	LSbj-1	6-8 26-30 per bartack	B	B
18.	Clean trousers.					
	a. Trim all thread ends.					
	b. Remove all loose threads.					
	c. Remove all spots and stains.					

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TABLE III - CONSTRUCTION OF ALUMINIZED COVER

No.	Operation	Stch Type	Seam/ Stch type	Stch in	Thread Ndl Bob/ Lpr
1.	Cutting.				
	a. Cut the cover in strict accordance with patterns which show directional lines, marks and notches for proper assembly. Directional lines shall not appear on the fabric. Drill holes are not permitted.				
	b. The component parts of the aluminized fabric may be cut from ends and when so cut shall finish with the aluminized side face up.				
	c. Cut 2 inch wide leather strips to bind the bottom edges of the leg openings.				
	d. Cut 15 inch lengths of elastic rubber tubing and connect ends together with tubing connectors.				
	e. The face side of the cover material is the aluminized side.				
	f. The face side of the lining material is the uncoated side.				
2.	Replacement of defective components.				
	During the spreading, cutting and manufacturing operations, components having material defects or damages that are classified as damages in MIL-STD-1666 shall be removed and replaced with nondefective and properly matched components.				
3.	Seam direction.				
	The lapping of the joining seams shall be as follows:				
	1. Fronts over back at side seams and inseams.				
	2. Left over right at seat seam and crotch seam.				

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TABLE III - CONSTRUCTION OF ALUMINIZED COVER

No.	Operation	Stch Type	Seam/ Stch type	Stch in	Thread Ndl	Thread Bob/ Lpr
4	Assemble leg protector.					
	a. Fold leg protector, end to end, face to face and stitch 3/8 to 7/16 inch from raw edges.	301	LSq-2(a)	6-8	B	B
	b. Turn leg protector to the finished position, turn seam allowance to one side and stitch 1/8 to 3/16 from the seam edge, securing the seam allowance.	301	LSq-2(b)	6-8	B	B
	c. Fold top edge of leg protector as indicated by the pattern marks, insert elastic leg tubing and stitch 1/2 inch from the folded edge enclosing the elastic in the leg protector. Care shall be taken not to catch the elastic in the stitching.	301	EFa-1	6-8	B	B
5.	Assemble aluminized trousers.	301	LSq-2	6-8	B	B
	a. Join the crotch seam, seat seam, outseam and inseam. Turn seam face side up and topstitch 1/8 to 3/16 inch from the turned edge.					
	b. Position waistband, face to face, at center back and stitch 3/8 to 1/2 inch from the raw edge.	301	SSa-1	6-8	B	B
	c. Position waistband on trouser, face to face, matching marks at side seams and the center back seams, stitch 3/8 inch from raw edges around waist.	301 or 401	SSe-2(a)	6-8	B	B
	d. Turn waistband to finished position and stitch 1/4 to 5/16 inch from the top folded edge of the trousers, thru all plies.	301	SSe-2(b)	6-8	B	B
	e. Stitch waist facing through all plies 1-3/4 inch from the finished top edge of waist facing.	301	similar to LSag-2	6-8	B	B

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TABLE III - CONSTRUCTION OF ALUMINIZED COVER

No.	Operation	Stch Type	Seam/ Stch type	Stch in	Thread Ndl	Thread Bob/ Lpr
5.	Assemble aluminized trousers. (Cont'd)					
	c. Bind the right and left fly openings in one continous operation with a 1-1/2 inch wide strip of alumininzed fabric. The fininshed side of the binding shall finish on the outside of the trouser. Binding shall finish even with the top edge of the trousers.	301	BSb-1	6-8	B	B
6.	Assemble and attach left fly extension.					
	a. Fold left fly extension piece face to face in accordance with the marks on the pattern. Stitch 3/8 (± 1/16) inch from the bottom and top raw edges.	301	SSe-1(a)	6-8	B	B
	b. Turn left fly extension piece to the finished positon and work out the corners. Stitch 3/16 (± 1/16) inch from the folded long edge and the turned top and bottom edges.	301	SSe-1(b)	6-8	B	B
	c. Positon a strip of 1 inch wide pile fastener tape on the length of the left fly extension as indicated by marks on the patterns. Stitch along all edges of the tape.	301	LSbj-1	6-8	B	B
	d. Position the raw edge of the left fly extension piece 1/2 inch past the bound edge of the left fly opening, with the wide end of the left fly extension piece even with the fininshed top edge of the trousers. Stitch 3/8 to 7/16 inch from the finished edge of the left fly opening thru all plies. The pile fastener tape shall finish on the inside of the trousers.	301	LSa-1	6-8	B	B
	e. Position a strip of hook fastener tape on the outside of the right front fly in accordance with template marker. Marker defines outer edge of tape and shall be positioned against front opening edge and top edge of finished aluminized cover as indicated. Stitch along top and bottom and side edges. The left fly shall overlap the right fly, with a smooth appearance when closed.	301	LSbj-1	6-8	B	B

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TABLE III - CONSTRUCTION OF ALUMINIZED COVER

No.	Operation	Stch Type	Seam/ Stch type	Stch in	Thread Ndl	Thread Bob/ Lpr
7.	Attach snap fasteners.					
	Center and attach six snap fastener sockets and caps to the waistband as indicated by pattern marker template (with the sockets facing the wearer). Template shall be carefully placed against the top and front edge (not left fly extension piece) on the finished aluminized cover.					
8.	Insert leg protector.					
	a. Insert leg protector into the leg, back to back matching joining seam with leg inseam and stitch 1/4 inch from the raw bottom edges.	301	SSa-1	6-8	B	B
	b. Doublestitch approximately 5-1/2 inches from the bottom raw edge according to the pattern marks. Stitching shall start and stop at the inseam.	301	SSv-2	6-8	B	B
9.	Bind leg bottoms.					
	a. Position a two inch wide leather strip onto the inside of the leg bottoms approximately 7/8 inch in from each leg bottom opening. Stitch 1/8 to 3/16 inch from the edge through all plies of material, overlapping ends of the strip approximately 1 to 1-1/2 inch near the inseam. Turn strip to the outside of the leg bottom and stitch 1/8 to 3/16 inch from the edge along free edge and across the overlapped portion through all plies of material.	301	LSa-1 (each edge)	6-8	B	B
	OR					
	b. As an option, fold leather strip in half lengthwise and using a folder, stitch each leg opening with a single or double row of stitching 1/8 to 3/16 inch from edges of leather strip, through all plies of material, overlapping ends of strip approximately 1 to 1-1/2 inches near the inseam. The ends of the legs shall extend within and to the folded edge of the leather strip.	301	BSa-1 or BSa-2	6-8	B	B

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TABLE III - CONSTRUCTION OF ALUMINIZED COVER

No.	Operation	Stch Type	Seam/ Stch type	Stch in	Thread Ndl	Thread Bob/ Lpr
10.	Labels.					
	a. Position and edgestitch the combination label, (see 3.3.13.3) on inside of trousers centered over seat seam with top edge of label 3/4 to 1 inch below bottom raw edge of waist facing.	301	LSbj-1	6-8	B	B
	b. Apply warning label, (see 3.3.13.4) horizontally across center back 2 to 3 inches below bottom edge of combination label.					
11.	Clean aluminized covers.					
	a. Trim all thread ends.					
	b. Remove all loose threads.					
	c. Remove all spots and stains.					
	NOTE: Care shall be taken in the use of solvents for cleaning to assure that the aluminized coating is not damaged.					

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3.8 Sizes and measurements. Sizes and measurements of the finished structural trousers and the aluminized proximity cover shall be as shown in Tables IV and V (see Figures 1 and 2). All measurements and tolerances are expressed in inches.

TABLE IV - STRUCTURAL TROUSERS

	A	B	C	
			INSEAMS	
SIZES	WAIST	SEAT	REGULAR	LONG
34	34	47	25	30
36	36	49	26	31
38	38	51	26	31
40	40	53	26	31
42	42	55	26	31
44	44	57	26	31
46	46	59	27	32
Tolerance	±1/2	±1/2	±1/2	±1/2

TABLE V - ALUMINIZED COVER

	A	B	C	
			INSEAMS	
SIZES	WAIST	SEAT	REGULAR	LONG
34	34	47	27	32
36	36	49	28	33
38	38	51	28	33
40	40	53	28	33
42	42	55	28	33
44	44	57	28	33
46	46	59	29	34
Tolerance	±1/2	±1/2	±1/2	±1/2

NOTE: The structural trousers and aluminized cover shall be laid flat and smooth with the front snap fasteners and fastener tapes closed.

Waist (A) - Double the measurement taken straight across the front of the structural trousers or proximity cover 3/4 inch down from top of waistband from folded edge to folded edge.

Seat (B) - Double the measurement taken straight across the front of the structural trouser or proximity cover from folded edge to folded edge, 1 inch above the bottom of fly.

Inseam (C) - The measurement shall be taken along the inseam from the junction of the front center seam to the bottom of the structural trousers or proximity cover.

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3.9 Workmanship. The finished structural trousers and aluminized cover shall conform to the quality established by this specification. The occurence of defects shall not exceed the acceptable quality levels.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the government to acceptance of defective material.

4.1.2 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

1. First article inspection (see 4.3).
2. Quality conformance inspection (see 4.4).

4.3 First article inspection. When required, the first article submitted in accordance with 3.2, shall be inspected as specified in 4.4.2 for compliance with design, construction, workmanship, appearance and dimensional requirements.

4.4 Quality conformance inspection. Sampling for inspection shall be performed in accordance with MIL-STD-1666, except where otherwise indicated.

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4.4.1 Component and material inspection. In accordance with 4.1 above, components and materials shall be tested in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable procurement documents. A certificate of compliance shall be furnished for paragraphs 3.3.6.1 - identification of silicone treatment. In addition, testing shall be performed on the components and characteristics listed in Table VI. The methods of testing specified in FED-STD-191, wherever applicable, and as listed in Table VI shall be followed. The physical and chemical values specified in Section 3, except where otherwise specified, apply to the average of the determinations made on the sample unit for test purposes as specified in the applicable test methods. All test reports shall contain the individual values utilized in expressing the final results. The lot shall be unacceptable if one or more sample units fail to meet any of the test requirements specified. The sample size shall be in accordance with the following:

LOT SIZE	SAMPLE SIZE
800 or less	2
801 thru 22,000	3
22,001 and over	5

The basis for lot size and sample units for testing shall be as follows:

COMPONENT	LOT SIZE EXPRESSION	SAMPLE UNIT
Laminated cloth	Yards	2 yards full width
Film only	Yards	1-1/2 yards full width
Leather	Square feet	1/2 square foot

TABLE VI - TESTING OF COMPONENTS

COMPONENT	CHARACTERISTIC	REQUIREMENT PARAGRAPH	TEST METHOD/ PARAGRAPH
Laminated cloth	Moisture vapor transmission	3.3.4	ASTM E96 procedure b 1/ 2/
	Hydrostatic resistance	3.3.4	5512 1/
	Water permeability after synthetic perspiration	3.3.4	5516 3/ 4/
Leather	Silicone identification	3.3.6.1	5/
	Water absorption	3.3.6.1	8111.1 of FED-STD-311 6/

1/ The film side shall face the water.

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2/ The free stream air velocity shall be 550 + 50 ft/min as measured 2 inches above the fabric specimen. The air flow shall be measured at least 2 inches away from any other surface. The test shall be run for 24 hrs. and weight measurements shall be taken at only the start and completion of the test. At the start of the 24 hour test period, the air gap between the water surface and the back of the specimen shall be 3/4 inch. Five specimens shall be tested.

3/ The water permeability shall be measured as specified in method 5516 of FED-STD-191, except that the face side of the laminated cloth shall contact the water. The hydrostatic head shall be 50 cm and shall be held for 3 minutes. The report shall only include measurement of the appearance of water drops. Leakage is defined as the appearance of water any place within the 4-1/2 inch diameter test area. The test may be performed using any test device which tests the same specimen area at the equivalent pressure using an appropriate size specimen. In cases of dispute, the apparatus described in method 5516 of FED-STD-191 shall be used.

4/ The synthetic perspiration solution shall be made up in a 500ml beaker by combining 3.0g sodium chloride, 1.0g trypticase soy broth powder, 1.0g normaetic stirring bar, and cover the beaker. Place the beaker on a combination hot plate/magnetic stirrer apparatus. While stirring, heat the solution to 50°c until all ingredients are dissolved. While stirring, cool the solution to 35°c, remove cover, and dispense immediately with a pipet or other suitable measuring device. Dispense 2 ml of perspiration solution at 35°c onto the center of an 8 x 8 inch glass. Place a specimen on the glass plate with the cloth side facing the glass. Dispense an additional 2 ml of the synthetic perspiration solution onto the center of the specimen. Place an 8 x 8 inch glass plate on top of the specimen with a 4 lb weight positioned in the center. Three specimens, 8 inches by 8 inches, shall be cut and exposed to perspiration as indicated above. After 16 hrs, remove the specimen (do not rinse) and test as specified in 3/.

5/ A certificate of compliance shall be acceptable for this characteristic.

6/ The leather shall be immerced for 30 minutes instead of the specified 24 hours.

4.4.2 Examination of the end item trouser and cover. Examination of the end item shall be in accordance with MIL-STD-1666 for Material Defects and Damages (Sec. I), Components and assembly (Sec. II), Cutting (Sec. III), Cleanness (Sec.IV), Seams and Stitchings (Sec. V), Metal Components (Sec VII), Dimensional examination specified in paragraph 4.3, Inspection levels and acceptable quality levels specified in paragraph 4.4, Examination of packaging requirements specified in 4.5 and in accordance with 4.4.2.1. Defects listed in paragraph 4.4.2.1 shall take precedence over defects listed for similar examination in MIL-STD-1666.

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4.4.2.1 Visual examination. The trousers and cover shall be examined for defects in construction, design and workmanship as follows:

NOTE: When an asterisk (*) appears, the defect shall be classified as follows:

MAJOR - When seriously affecting serviceability or appearance.
MINOR - When affecting serviceability or appearance, but not seriously.

CLASSIFICATION OF DEFECTS

DEFECT	CLASSIFICATION MAJOR	MINOR
LEATHER		
a. Loose, spongy or flanky	X	
b. Any open vein	X	
c. Any rupture		X
d. Not soft or pliable		X
e. Any scar, brittle or bony area, weak or thin spot or abraded area	X	
REFLECTIVE TAPE		
a. One section missing		X
b. More than one section missing	X	
c. Edge of tape not parallel with bottom edge of trouser legs by more than 1/2 inch		X
FASTENER TAPE		
a. Any section missing	*	
b. Any section incorrect component, i.e. hook portion instead of pile portion or pile portion instead of hook portion	*	
c. Not edge stitched along all edges when specified		X
WEBBING		
a. Not tightly woven		X
b. Raveled edge		X
TOP EDGE OF TROUSERS		
a. Excessive fullness, twist or puckering at joining to waist facing		X
b. Waist facing exposed beyond top edge of trousers (score only when the condition exists on more than 1/2 the length of the seam)		X

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DEFECT	CLASSIFICATION MAJOR	MINOR
SUSPENDERS		
a. Straps crossed at a distance less than 5 inches or more than 5-1/2 inches when measured from finished top edge of waist along center of straps to center of diamond stitch		X
b. Any strap misplaced, attached in other than specified manner, or insecurely attached	X	
c. Straps not crossed at a 40 to 50° angle		X
d. Finished length of strap less than 39-1/2 inches or more than 40-1/2 inches		X
FLY CONSTRUCTION AND CLOSURE		
a. Waist edges not even by more than 3/8 inch when fly is fastened		X
b. Sockets and studs out of alignment causing a noticeable bulge or twist on front fly when fastened		X
c. Fly base noticeably twisted or puckered		X
LEATHER KNEE PATCHES		
a. Leather knee patch missing		X
b. Leather knee patch misplaced or not attached as specified		X
LEG BOTTOMS (aluminized cover only)		
a. Bottom of legs not leather bound		X
b. Leather binding on legs overlapped less than 1 inch or more than 1-1/2 inch		X
LEG PROTECTORS (aluminized cover only)		
a. Leg protector missing	X	
b. Leg protector misplaced or not attached as specified		X
c. Elastic tubing caught in stitching		X
LEG TABS (structural trouser only)		
a. Leg tab missing		X
b. Leg tab less than 1 inch wide		X
c. Leg tab less than 4-1/2 or more than 5 inches long		X
d. Not attached or positioned as specified		X

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DEFECT	CLASSIFICATION MAJOR	MINOR
HEM (structural trouser only)		
a. Width of hem less than 1-3/8 or more than 1-5/8 inches		X
b. Not uniform in width by more than 1/4 inch		X
LABEL		
a. Omitted, one or more		X
b. Not attached as specified		X
c. Inscription or instruction omitted, incomplete, illegible or incorrect		X
d. Position not as specified		X
INSTRUCTION MANUAL		
a. Omitted or not positioned as specified		X

4.4.4 Palletization examination. An examination shall be made to determine that the palletization complies with the section 5 requirements. Defects shall be scored in accordance with the list below. The sample unit shall be one palletized unit load fully packaged. The lot size shall be the number of palletized unit loads in the end item inspection lot. The inspection level shall be S-1 and the AQL, expressed in terms of defects per hundred units, shall be 6.5 as specified in MIL-STD-105.

Examine	Defect
Finished dimension	Length, width, or height exceeds specified maximum requirements.
Palletization	Pallet pattern not as specified. Interlocking of loads not as specified. Load not bonded with required straps as specified.
Weight	Exceeds maximum load limits.
Marking	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application.

5. PACKAGING

5.1 Preservation packaging. Preservation-packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A.

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5.1.1.1 Structural trousers and aluminized covers. Structural trousers and aluminized covers shall be folded as follows:

1. Close front fly.
2. Place fastened suspenders inside trousers (Structural trouser only).
3. Fold trousers in half lengthwise with the fly closure on the outside of the fold
4. Fold in half across the length by bringing the bottom of the trousers up to the top.

Folded structural trousers and aluminized covers shall measure approximately 23 inches by 15 inches. Each folded trouser shall be individually packaged in a snug-fitting, clear polyethylene film bag having a film thickness of not less than 0.003 inch. The bag may be fabricated from polyethylene film tubing or sheeting. All seams and closures shall be formed by heat sealing. A 1/4 inch diameter hole shall be made at one corner of each bag to allow excess air to escape. As an alternate, the polytheylene bag may be of the tuck in and reverse flap type, in which a heat seal closure and corner hole are not required. The instruction manual must be inserted into the polyethylene bag and positioned so that it is visible.

5.1.2 Level C (Commercial packaging). Structural trousers and aluminized covers shall be packaged separately to afford adequate protection against physical damage during shipment from the supplier to the first receiving activity. The package and the quantity per package shall be the same as that normally used by the supplier for retail distribution.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Structural trousers. Fifteen structural trousers, of one size only, packaged as specified in 5.1, shall be packed in a fiberboard shipping container assembled, closed and reinforced conforming to type CF, class weather-resistant, grade V15c, variety DW, size 3A of MIL-B-17757. The fiberboard used for the liner shall conform to type CF, class domestic conforming to MIL-B-17757. Level A folded trousers shall be stacked flat with each pair of trouser reversed end to end, fifteen in depth within the shipping container. The length of the trouser shall be parallel to the side panels of the shipping container.

5.2.1.2 Aluminized covers. Twenty covers of one size only, packaged as specified in 5.1, shall be packed in a fiberboard shipping container assembled, closed and reinforced conforming to type CF, class weather-resistant, grade V15c, variety DW, size 3A of MIL-B-17757. The fiberboard used for the liner shall conform to type CF, class domestic conforming to MIL-B-17757. Level A folded covers shall be stacked on edge within the shipping container with the face the package parallel to the end panels within the shipping container.

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5.2.2 Level B.

5.2.2.1 Structural trousers. Fifteen structural trousers of one size only, packaged as specified in 5.1, shall be packed in a fiberboard shipping container assembled, closed and reinforced conforming to type CF, class domestic, grade 275, variety DW, size 3A of MIL-B-17757. The fiberboard used for the liner shall conform to type CF, class domestic conforming to MIL-B-17757. Level A folded trousers shall be stacked flat with each pair of trousers reversed end to end, fifteen in depth within the shipping container. The length of the trousers shall be parallel to the side panels of the shipping container.

5.2.2.2 Aluminized cover. Twenty covers of one size only, packaged as specified in 5.1, shall be packed in a fiberboard shipping container assembled, closed and reinforced conforming to type CF, class domestic, grade 275, variety DW, size 3A of MIL-B-17757. The fiberboard used for the liner shall conform to type CF, class domestic conforming to MIL-B-17757. Level A folded covers shall be stacked on edge within the shipping container with the face of the package parallel to the end panels within the shipping container.

5.2.3 Level C (Commercial packing). Structural trousers and aluminized covers separately packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. The quantity per shipping container shall be the same as that normally used by the supplier for retail distribution. Containers shall comply with U.S Postal Service Manual, Uniform Freight Classification Rules or National Motor Freight Classification Rules, as applicable.

5.3 Palletization. When specified (see 6.2), item, packed as specified in 5.2, shall be palletized on a 4-way entry pallet in accordance with load type Ia of MIL-STD-147. Each prepared load shall be banded with primary and secondary straps in accordance with the banding means C, K and L, or O or P. Pallet patterns shall be in accordance with the appendix of MIL-STD-147.
The pallet shall be 4-way, Type IV; Type V, class 1, size 2; or Type VIII, fabricated from wood groups I, II, III or IV, Grade A of NN-P-71, or 4-way, Style 1, Size A, Type I, Class 1 fabricated from groups specified of MIL-P-15011. Interlocking of loads shall be effected by reversing the pattern of each course. If the container is of a size which does not conform to any of the patterns specified in MIL-STD-147, the pallet pattern used shall first be approved by the contacting officer.

5.4 Marking. In addition to any special marking required by the contract or order, interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

5.4.1 Polyethylene bagged packages. Polyethylene bagged packages shall have the required identification information legibly printed or stamped in black directly on the bag across the center face or on a white paper label inserted within the bag so as to permit ready identification. The bag or label shall only indicate the following information:

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STOCK NUMBER
NOMENCLATURE
SIZE
QUANTITY

6. NOTES

6.1 Intended use. The trousers are intended for wear by DOD personnel engaged in firefighting. The structural trouser is for use by firefighters engaged in structural type fires. The structural trouser with the aluminized cover attached is for use by firefighters engaged in proximity type fire fighting.

6.2 Ordering data. Procurement documents should specify the following:

a. Title and date of this specification.
b. Types, classes, sizes and lengths required (see 1.2)
c. When first article inspection is required (see 3.2), the item will be tested and should be a first article sample. The contracting officer should include specific instructions in acquistion documents regarding arrangements for examination, quantity and testing and approval of the first article.
d. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
e. List of approved silicone compounds (see 3.3.6.1).
f. When palletization is required (see 5.3).

6.3 Samples and patterns. For access to samples and patterns, address the procuring activity issuing the invitation for bids.

6.4 Recycled materials. It is encouraged that recycled materials be used when practical as long as it meets the requirements of this specification.

6.5 A suggested source of supply for the laminated material specfied in 3.3.4 is W. L. Gore and Associates, Inc., 3 Blue Ball Rd., P.O. Box 1130, Elkton, Maryland 21921.

6.6 A suggested source of supply for the reflective tape specified in 3.3.7 is 3M Safety and Security Systems Division, St. Paul, Minnesota 55144.

6.7 A suggested sorce of supply for the tubing connectors specified in 3.3.8.2 is McMaster Carr Supply Co., 600-T County Line Road, Elmhurst, IL 60126.

6.8 A suggested source of supply for the combination stud and socket is Scovill Inc., 1100-T Buckingham St.., Watertown, CT 06795.

6.9 Silicone compounds and flame resistant treatments. Only those silicone compounds and flame resistant treatments approved by the U.S. Army Natick Research, Development and Engineering Center, Natick, Massachusetts 01760, and the appropriate medical service and so listed in the invitation for bids or request for proposal, shall be considered acceptable for the related procurement (see 3.3.6.1).

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6.10 Equal item. Prior to the use of an "or equal" item, the contractor shall submit the item with supporting data to the contracting officer for subsequent approval or disapproval by the responsible Military Agency.

6.11 Subject term (key word) listing.
Cloth, para-aramid/PBI
Cloth, aluminized
Clothing, protective
Trouser, combination
Trouser, proximity
Coat, structural

6.12 Liner. The liner mentioned in the warning labels specified in paragraphs 3.3.13.1 and 3.3.13.2 refers to the liner covered by specification; MIL-L-24942 - Liner, Fireman's Trouser, Quilted.

Custodians:
Navy - NU
Army - GL
Air Force - 99

Review activities:
Navy - MC
Army - MD
Air Force - 82
DLA - CT

User activities:
Navy - SH, CG
Air Force - 45

Preparing activity:
Navy - NU

Project No. 8415-0630

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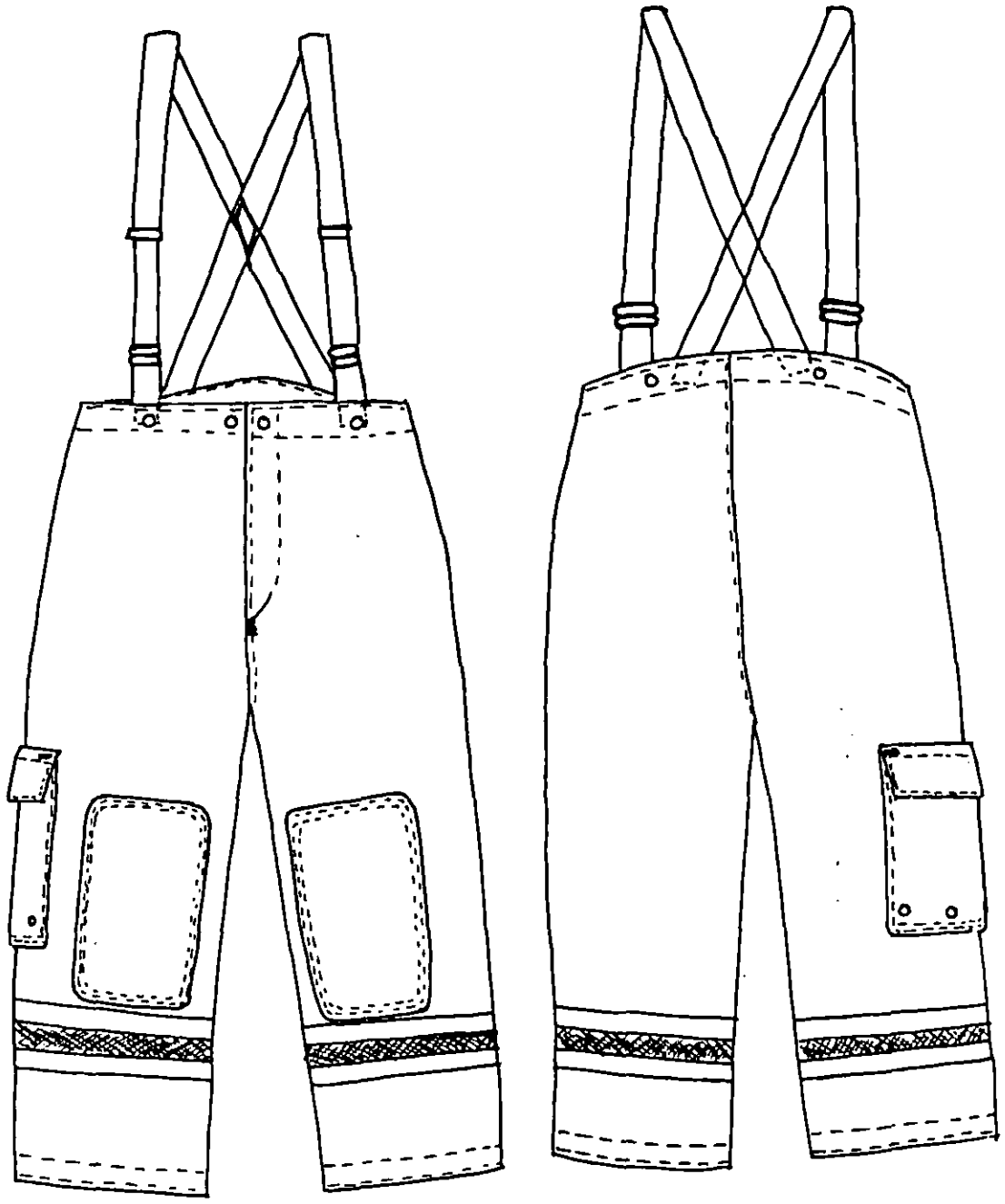
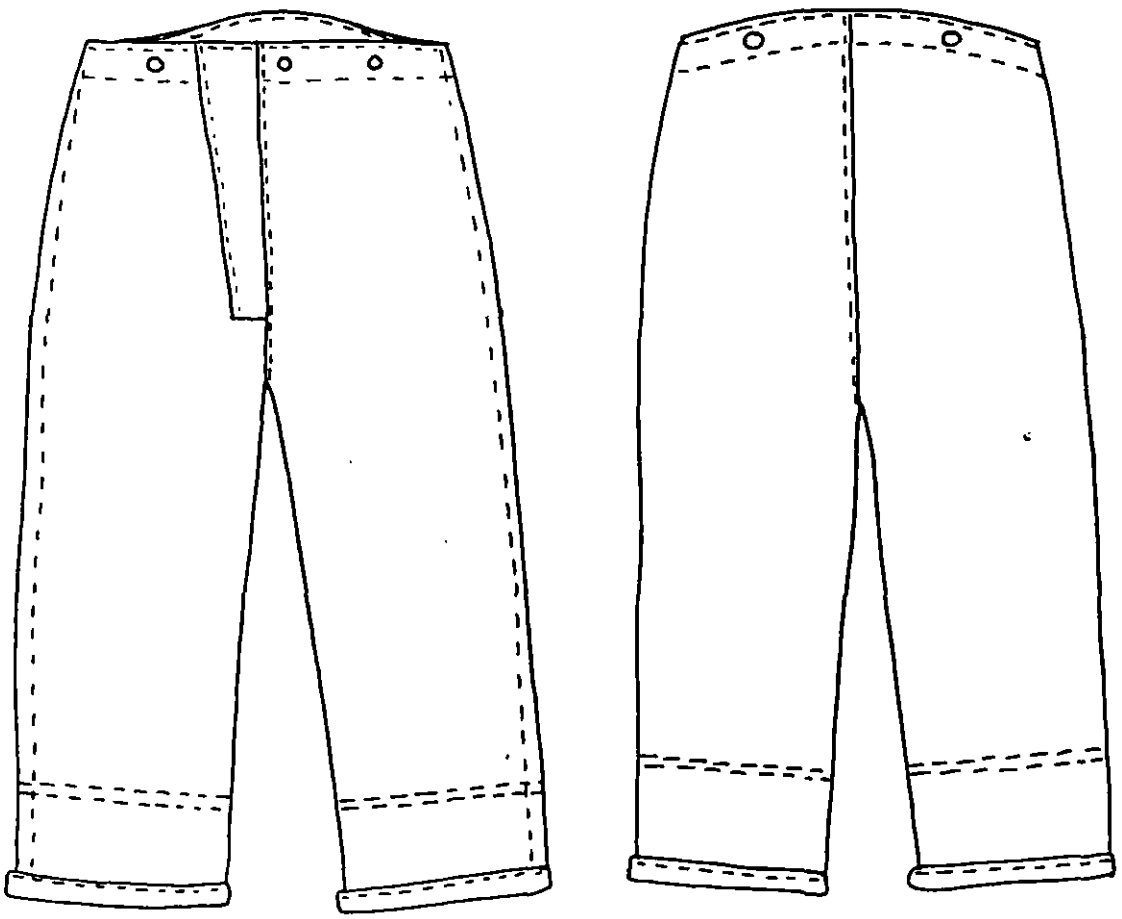


FIGURE - 1

STRUCITURAL TROUSERS

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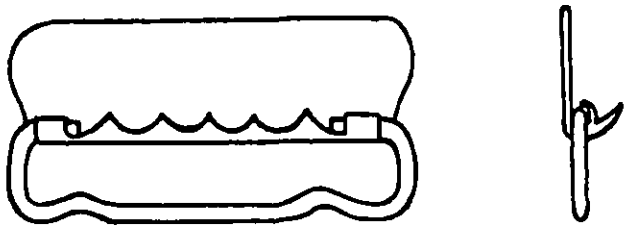


FRONT BACK

FIGURE - 2

COVER, FIREMEN'S, TROUSERS (PROXIMITY)

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BUCKLE, CLAMP TYPE

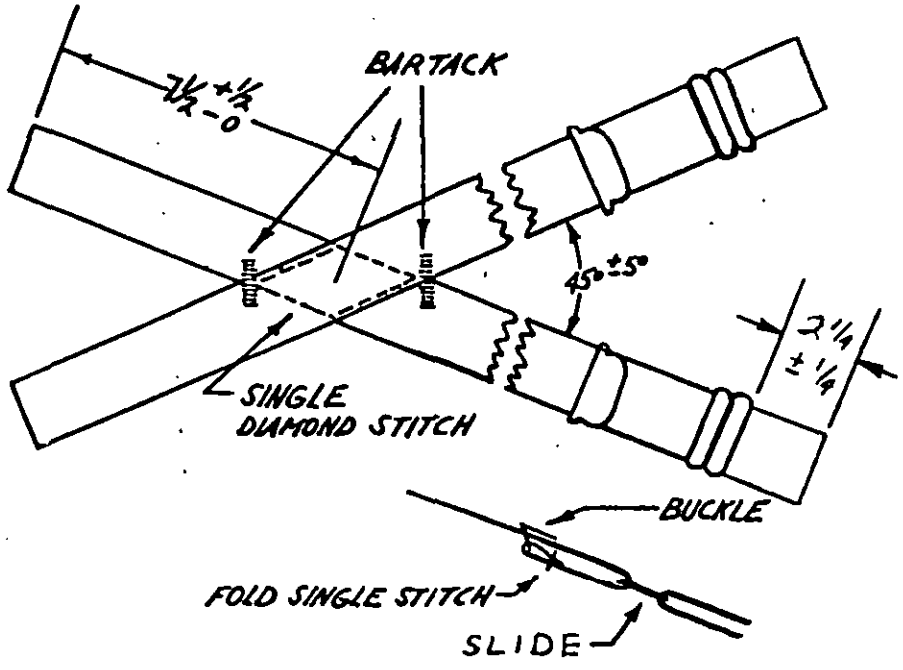


FIGURE 3 - SUSPENDER ASSEMBLY

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-T-24940	TROUSERS, FIREMEN'S, STRUCTURAL/ALUMINIZED PROXIMITY

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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